# FOLLOWUP-selector-extra-syzygies (follow-up to LEAD-A-selector-eta-kernel), 2026-11-12

All result statements below are [pcc:unverified].

## Preregistration (written before any computation in this directory)

Frozen at 2026-11-12, before syz.c was written or run.

**Object.** For an unsatisfiable instance (n, k = (n+1)/2, V, z, a6) and a degree e, K_e is the kernel of the homogeneous selector system (all 2^k points, rows Tr(lambda(C) Phi*_C(v_i)), Tr(lambda(C) h_C), unknowns the ANF coefficients of lambda: V -> GF(2^n) of degree <= e). T_e = { q (1 + ell_0) zeta : deg q <= e-1 }, zeta = z^-2, ell_0(C) = Tr(C) + Tr(a6/z^2) (lane_A_falldegree.md, Lead A1, F3), dim T_e = M(k-1, e-1). The **excess** is x_e = dim K_e - dim T_e, computed only after checking T_e <=(sub) K_e.

**Prediction P (from the task card).** The excess is nonzero only at e = e_min.

Precise reading, fixed now: because K_e <=(sub) K_{e+1} and T_{e+1} cap (deg <= e) = T_e, a nonzero excess at e_min forces x_e >= x_{e_min} > 0 for every e > e_min. So P is only testable at e <= e_min and is read as:

- P1: x_e = 0 for every tested e < e_min;
- P2: x_{e_min} > 0 for every tested instance.

**Cells** (instances: the committed unsatisfiable seeds of lfd_small.make_instance, e_min from out/c/emin_n*.jsonl):

| n | k | e_min | seeds | e tested |
|---|---|---|---|---|
| 15 | 8 | 3 | 1, 5, 6 | 2, 3 |
| 17 | 9 | 3 | 1, 2, 8 | 2, 3 |
| 19 | 10 | 4 | 4, 7, 10 | 3, 4 |
| 21 | 11 | 4 | 3, 11 | 3, 4 |
| 23 | 12 | 4 | 4, 8, 11 | 3 (control), 4 |
| 25 | 13 | 4 | 5, 6, 7 | 3 (control), 4 |
| 27 | 14 | 5 | 3, 6 | 4 (control, already known 0), 5 if it fits in 3 GB / 20 min |

**Falsifiers.** P is falsified by any cell with x_e > 0 at e < e_min (P1), or any instance with x_{e_min} = 0 (P2). Any instance with T_e not inside K_e, or rank T_e != M(k-1,e-1), or a kernel vector that fails the independent point-evaluation check, is a bug and voids the cell.

**Controls.** - Reproduction: the kernel dimension at n = 23, 25 must equal the committed nullity_grid.jsonl values (1144/1145/1143, 741/740/740) to the digit. - Independent check: every kernel basis vector is evaluated on all 2^k points by a Moebius transform and every homogeneous row is re-tested by parity (code path independent of the echelon form). - Trivial family: T_e <=(sub) K_e, rank T_e = M(k-1,e-1).

**Characterisation (descriptive, no thresholds; reported as found):** degree filtration of the excess, support on the two halves H_0 = {ell_0 = 0}, H_1 = {ell_0 = 1}, fibre ranks against the pointwise annihilator dimension, the V-independent ("universal") subfamily {lambda : lambda(C) perp Phi*_C(GF(2^n)) and perp h_C for all C}, and closure under Frobenius (coefficientwise squaring) and under multiplication by the field generator x.

### Addendum to the preregistration (written 06:46 UTC, after n = 15-27 seed 3 data, before the n = 27 seed 6, e = 5 run)

Post-hoc observation from the first 36 cells: at e = e_min the kernel dimension equals a per-n base plus h, the number of points C in V with C = X1 + X2, X1, X2 in V, X1 X2 = z. Out-of-sample prediction for n = 27, seed 6 (h = 2), e = 5: kernel dimension 10400 + 1 = 10401 (seed 3 has h = 1 and 10400). Falsified by any other value.

Result of the out-of-sample check: n = 27, seed 6, e = 5 gave kernel dimension **10401**, as predicted ([pcc:unverified]).

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## What was run

All commands are run from `research/lfd_lower_bound_20261111/`. Each job ran single-threaded. `emin.c` is not modified (`git diff` on it is empty), and every new file is in this directory.

- **Tool.** `out/lead_A1_syzygies/syz.c`, compiled with `gcc -O2 -march=native -o syz syz.c -lm4ri -lm`. It has three modes:
  - `syz E PERMILLE SEED`: the kernel analysis.
    - It builds the homogeneous rows of emin.c (same row builder, same column layout).
    - It extracts a kernel basis from the reduced echelon form (M4RI).
    - It cuts that basis to the exact kernel by evaluating every vector on all `2^k` points. This step only matters when `PERMILLE < 1000` (a point sample).
    - It re-checks the final basis by Moebius evaluation and per-row parity on all `2^k (k+1)` rows.
    - It builds `T_e` from the F3 formula and checks `T_e <=(sub) K_e` the same way.
    - It reports ranks, the degree filtration, support on `H_0`/`H_1`, the V-independent subfamily, fibre ranks against `dim A_C`, and closure under coefficientwise squaring and under multiplication by `x`.
  - `syz dual D`: the dual section count `dim S_D` (see the reading below).
  - `syz coset H E`: the sections over one half `H_H` only, in its `k - 1` coordinates.
- **Main grid** (`grid.jsonl`, `run.log`):
  - `out/lead_A1_syzygies/run_grid.sh "15:1,5,6:2,3" "17:1,2,8:2,3" "19:4,7,10:3,4" "21:3,11:3,4"`
  - `out/lead_A1_syzygies/run_grid.sh "23:4,8,11:3,4" "25:5,6,7:3,4"`
  - `out/lead_A1_syzygies/run_grid.sh "27:3:4" "27:3:5:460"`
  - `out/lead_A1_syzygies/run_grid.sh "27:6:5:460"`
  - The `n = 27, e = 5` runs use a sample of 7,536 of the 16,384 points. The sample kernel had violation rank 0 on all points, so it is the exact kernel. The cost was 1.53 GB peak and 744 s / 761 s.
- **Post-hoc tools:**
  - `high_points.py N SEED` (`high_points.jsonl`) gives `h`.
  - `syz dual D` at `D = k - e - 1` (`dual.jsonl`) covers 20 cells at n = 15-23.
  - `syz coset H E` (`coset.jsonl`) covers 46 half-cells.
- **Tables, figure and PDF:** `RESULT.pdf` is a pandoc render of this file, with Unicode symbols spelled out in ASCII. `summarize.py` writes `tables.md` and `plot.py` writes `kernel_vs_trivial.svg` and `kernel_vs_trivial.png`.

## Controls

- **Reproduction:** passes. The kernel dimension at `n = 23` (1144 / 1145 / 1143) and `n = 25` (741 / 740 / 740) equals the committed `out/lead_A1/nullity_grid.jsonl` to the digit, and so does `n = 27, e = 4` (378).
- **Independent evaluation check:** every final kernel basis vector passes in all 37 cells (`kernel_check_ok`).
- **Trivial family:** in all 37 cells, `T_e <=(sub) K_e` and `rank T_e = M(k-1, e-1)`. F1/F2 hold at every point (`F1F2_violations = 0`).
- **Default unchanged after adding the dual/coset modes:** the rebuilt `syz` reproduces the grid JSON of `n = 19` seed 4 and `n = 23` seed 4 at `e = 4` field for field (only time and RSS differ).

## Results [pcc:unverified]

### Excess per cell, and the preregistered test

`h` is the number of `C` in `V` with `C = X1 + X2`, `X1, X2 in V`, `X1 X2 = z`.

| n | k | seed | e | e_min | kernel dim | M(k-1,e-1) = rank T | T in K | excess | P | h | cols - E_eff | dim S_{k-e-1} (dual) | identity |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | 8 | 1 | 2 | 3 | 8 | 8 | yes | 0 | P1 ok | 0 | -120 | 1502 | yes |
| 15 | 8 | 1 | 3 | 3 | 146 | 29 | yes | 117 | P2 ok | 0 | 146 | 800 | yes |
| 15 | 8 | 5 | 2 | 3 | 8 | 8 | yes | 0 | P1 ok | 0 | -120 | - | - |
| 15 | 8 | 5 | 3 | 3 | 146 | 29 | yes | 117 | P2 ok | 0 | 146 | - | - |
| 15 | 8 | 6 | 2 | 3 | 8 | 8 | yes | 0 | P1 ok | 1 | -120 | 1501 | yes |
| 15 | 8 | 6 | 3 | 3 | 147 | 29 | yes | 118 | P2 ok | 1 | 146 | 800 | yes |
| 17 | 9 | 1 | 2 | 3 | 9 | 9 | yes | 0 | P1 ok | 0 | -263 | 3836 | yes |

| n | k | seed | e | e_min | kernel dim | M(k-1,e-1) = rank T | T in K | excess | P | h | cols - E_eff | dim S_{k-e-1} (dual) | identity |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 17 | 9 | 1 | 3 | 3 | 87 | 37 | yes | 50 | P2 ok | 0 | 31 | 2486 | yes |
| 17 | 9 | 2 | 2 | 3 | 9 | 9 | yes | 0 | P1 ok | 0 | -263 | - | - |
| 17 | 9 | 2 | 3 | 3 | 87 | 37 | yes | 50 | P2 ok | 0 | 31 | - | - |
| 17 | 9 | 8 | 2 | 3 | 9 | 9 | yes | 0 | P1 ok | 1 | -263 | 3835 | yes |
| 17 | 9 | 8 | 3 | 3 | 88 | 37 | yes | 51 | P2 ok | 1 | 31 | 2486 | yes |
| 19 | 10 | 4 | 3 | 4 | 46 | 46 | yes | 0 | P1 ok | 1 | -265 | 6942 | yes |
| 19 | 10 | 4 | 4 | 4 | 1164 | 130 | yes | 1034 | P2 ok | 1 | 1163 | 4070 | yes |
| 19 | 10 | 7 | 3 | 4 | 46 | 46 | yes | 0 | P1 ok | 2 | -265 | 6941 | yes |
| 19 | 10 | 7 | 4 | 4 | 1165 | 130 | yes | 1035 | P2 ok | 2 | 1163 | 4070 | yes |
| 19 | 10 | 10 | 3 | 4 | 46 | 46 | yes | 0 | P1 ok | 1 | -265 | - | - |
| 19 | 10 | 10 | 4 | 4 | 1164 | 130 | yes | 1034 | P2 ok | 1 | 1163 | - | - |
| 21 | 11 | 3 | 3 | 4 | 56 | 56 | yes | 0 | P1 ok | 2 | -849 | 17711 | yes |
| 21 | 11 | 3 | 4 | 4 | 1211 | 176 | yes | 1035 | P2 ok | 2 | 999 | 11936 | yes |
| 21 | 11 | 11 | 3 | 4 | 56 | 56 | yes | 0 | P1 ok | 1 | -849 | 17712 | yes |
| 21 | 11 | 11 | 4 | 4 | 1210 | 176 | yes | 1034 | P2 ok | 1 | 999 | 11936 | yes |
| 23 | 12 | 4 | 3 | 4 | 67 | 67 | yes | 0 | P1 ok | 1 | -1860 | 42342 | yes |
| 23 | 12 | 4 | 4 | 4 | 1144 | 232 | yes | 912 | P2 ok | 1 | 153 | 32034 | yes |
| 23 | 12 | 8 | 3 | 4 | 67 | 67 | yes | 0 | P1 ok | 2 | -1860 | - | - |
| 23 | 12 | 8 | 4 | 4 | 1145 | 232 | yes | 913 | P2 ok | 2 | 153 | 32034 | yes |
| 23 | 12 | 11 | 3 | 4 | 67 | 67 | yes | 0 | P1 ok | 0 | -1860 | - | - |
| 23 | 12 | 11 | 4 | 4 | 1143 | 232 | yes | 911 | P2 ok | 0 | 153 | 32034 | yes |
| 25 | 13 | 5 | 3 | 4 | 79 | 79 | yes | 0 | P1 ok | 1 | -3473 | - | - |
| 25 | 13 | 5 | 4 | 4 | 741 | 299 | yes | 442 | P2 ok | 1 | -1900 | - | - |
| 25 | 13 | 6 | 3 | 4 | 79 | 79 | yes | 0 | P1 ok | 0 | -3473 | - | - |
| 25 | 13 | 6 | 4 | 4 | 740 | 299 | yes | 441 | P2 ok | 0 | -1900 | - | - |
| 25 | 13 | 7 | 3 | 4 | 79 | 79 | yes | 0 | P1 ok | 0 | -3473 | - | - |
| 25 | 13 | 7 | 4 | 4 | 740 | 299 | yes | 441 | P2 ok | 0 | -1900 | - | - |
| 27 | 14 | 3 | 4 | 5 | 378 | 378 | yes | 0 | P1 ok | 1 | -5903 | - | - |
| 27 | 14 | 3 | 5 | 5 | 10400 | 1093 | yes | 9307 | P2 ok | 1 | 6538 | - | - |
| 27 | 14 | 6 | 5 | 5 | 10401 | 1093 | yes | 9308 | P2 ok | 2 | 6538 | - | - |

**Characterisation per cell**

| n | seed | e | degree filtration [d, dim K_<=d, dim T_<=d] (last two) | rank on H1 | dim vanishing on H1 | universal | Frobenius-closed | x-closed | fibre rank = dim A_C at every point |
|---|---|---|---|---|---|---|---|---|---|
| 15 | 1 | 2 | [[1, 1, 1], [2, 8, 8]] | 0 | 8 | 8 | 0 | 0 | no (0 on all of H1) |
| 15 | 1 | 3 | [[2, 8, 8], [3, 146, 29]] | 117 | 29 | 29 | 0 | 0 | yes |
| 15 | 5 | 2 | [[1, 1, 1], [2, 8, 8]] | 0 | 8 | 8 | 0 | 0 | no (0 on all of H1) |
| 15 | 5 | 3 | [[2, 8, 8], [3, 146, 29]] | 117 | 29 | 29 | 0 | 0 | yes |
| 15 | 6 | 2 | [[1, 1, 1], [2, 8, 8]] | 0 | 8 | 8 | 0 | 0 | no (0 on all of H1) |

| n | seed | e | degree filtration [d, dim K_<=d, dim T_<=d] (last two) | rank on H1 | dim vanishing on H1 | universal | Frobenius-closed | x-closed | fibre rank = dim A_C at every point |
|---|---|---|---|---|---|---|---|---|---|
| 15 | 6 | 3 | [[2, 8, 8], [3, 147, 29]] | 118 | 29 | 29 | 0 | 0 | yes |
| 17 | 1 | 2 | [[1, 1, 1], [2, 9, 9]] | 0 | 9 | 9 | 0 | 0 | no (0 on all of H1) |
| 17 | 1 | 3 | [[2, 9, 9], [3, 87, 37]] | 50 | 37 | 37 | 0 | 0 | yes |
| 17 | 2 | 2 | [[1, 1, 1], [2, 9, 9]] | 0 | 9 | 9 | 0 | 0 | no (0 on all of H1) |
| 17 | 2 | 3 | [[2, 9, 9], [3, 87, 37]] | 50 | 37 | 37 | 0 | 0 | yes |
| 17 | 8 | 2 | [[1, 1, 1], [2, 9, 9]] | 0 | 9 | 9 | 0 | 0 | no (0 on all of H1) |
| 17 | 8 | 3 | [[2, 9, 9], [3, 88, 37]] | 51 | 37 | 37 | 0 | 0 | yes |
| 19 | 4 | 3 | [[2, 10, 10], [3, 46, 46]] | 0 | 46 | 46 | 0 | 0 | no (0 on all of H1) |
| 19 | 4 | 4 | [[3, 46, 46], [4, 1164, 130]] | 1034 | 130 | 130 | 0 | 0 | yes |
| 19 | 7 | 3 | [[2, 10, 10], [3, 46, 46]] | 0 | 46 | 46 | 0 | 0 | no (0 on all of H1) |
| 19 | 7 | 4 | [[3, 46, 46], [4, 1165, 130]] | 1035 | 130 | 130 | 0 | 0 | yes |
| 19 | 10 | 3 | [[2, 10, 10], [3, 46, 46]] | 0 | 46 | 46 | 0 | 0 | no (0 on all of H1) |
| 19 | 10 | 4 | [[3, 46, 46], [4, 1164, 130]] | 1034 | 130 | 130 | 0 | 0 | yes |
| 21 | 3 | 3 | [[2, 11, 11], [3, 56, 56]] | 0 | 56 | 56 | 0 | 0 | no (0 on all of H1) |
| 21 | 3 | 4 | [[3, 56, 56], [4, 1211, 176]] | 1035 | 176 | 176 | 0 | 0 | yes |
| 21 | 11 | 3 | [[2, 11, 11], [3, 56, 56]] | 0 | 56 | 56 | 0 | 0 | no (0 on all of H1) |

| n | seed | e | degree filtration [d, dim K_<=d, dim T_<=d] (last two) | rank on H1 | dim vanishing on H1 | universal | Frobenius-closed | x-closed | fibre rank = dim A_C at every point |
|---|---|---|---|---|---|---|---|---|---|
| 21 | 11 | 4 | [[3, 56, 56], [4, 1210, 176]] | 1034 | 176 | 176 | 0 | 0 | yes |
| 23 | 4 | 3 | [[2, 12, 12], [3, 67, 67]] | 0 | 67 | 67 | 0 | 0 | no (0 on all of H1) |
| 23 | 4 | 4 | [[3, 67, 67], [4, 1144, 232]] | 912 | 232 | 232 | 0 | 0 | yes |
| 23 | 8 | 3 | [[2, 12, 12], [3, 67, 67]] | 0 | 67 | 67 | 0 | 0 | no (0 on all of H1) |
| 23 | 8 | 4 | [[3, 67, 67], [4, 1145, 232]] | 913 | 232 | 232 | 0 | 0 | yes |
| 23 | 11 | 3 | [[2, 12, 12], [3, 67, 67]] | 0 | 67 | 67 | 0 | 0 | no (0 on all of H1) |
| 23 | 11 | 4 | [[3, 67, 67], [4, 1143, 232]] | 911 | 232 | 232 | 0 | 0 | yes |
| 25 | 5 | 3 | [[2, 13, 13], [3, 79, 79]] | 0 | 79 | 79 | 0 | 0 | no (0 on all of H1) |
| 25 | 5 | 4 | [[3, 79, 79], [4, 741, 299]] | 442 | 299 | 299 | 0 | 0 | yes |
| 25 | 6 | 3 | [[2, 13, 13], [3, 79, 79]] | 0 | 79 | 79 | 0 | 0 | no (0 on all of H1) |
| 25 | 6 | 4 | [[3, 79, 79], [4, 740, 299]] | 441 | 299 | 299 | 0 | 0 | yes |
| 25 | 7 | 3 | [[2, 13, 13], [3, 79, 79]] | 0 | 79 | 79 | 0 | 0 | no (0 on all of H1) |
| 25 | 7 | 4 | [[3, 79, 79], [4, 740, 299]] | 441 | 299 | 299 | 0 | 0 | yes |
| 27 | 3 | 4 | [[3, 92, 92], [4, 378, 378]] | 0 | 378 | 378 | 0 | 0 | no (0 on all of H1) |
| 27 | 3 | 5 | [[4, 378, 378], [5, 10400, 1093]] | 9307 | 1093 | 1093 | 0 | 0 | yes |

| n | seed | e | degree filtration [d, dim K_<=d, dim T_<=d] (last two) | rank on H1 | dim vanishing on H1 | universal | Frobenius-closed | x-closed | fibre rank = dim A_C at every point |
|---|---|---|---|---|---|---|---|---|---|
| 27 | 6 | 5 | [[4, 378, 378], [5, 10401, 1093]] | 9308 | 1093 | 1093 | 0 | 0 | yes |

The `cols - E_eff` column uses the lead's rank bound `E_eff = (k+1)M(k,e+1) - M(k,e+2) + 1`.

**Kernel dimensions at e_min, with bases.** The base is the excess minus `h`.

| n | k | e_min | kernel dim at e_min | excess | excess - h (base) |
|---|---|---|---|---|---|
| 15 | 8 | 3 | 146, 146, 147 | 117, 117, 118 | 117 |
| 17 | 9 | 3 | 87, 87, 88 | 50, 50, 51 | 50 |
| 19 | 10 | 4 | 1164, 1165, 1164 | 1034, 1035, 1034 | 1033 |
| 21 | 11 | 4 | 1211, 1210 | 1035, 1034 | 1033 |
| 23 | 12 | 4 | 1144, 1145, 1143 | 912, 913, 911 | 911 |
| 25 | 13 | 4 | 741, 740, 740 | 442, 441, 441 | 441 |
| 27 | 14 | 5 | 10400, 10401 | 9307, 9308 | 9306 |

## Verdict against the preregistered falsifiers

**P holds in every tested cell** `[pcc:unverified]`.

- **P1:** the excess is 0 at all 18 instances with `e = e_min - 1`: n = 15, 17 at e = 2; n = 19, 21, 23, 25 at e = 3; n = 27 at e = 4.
- **P2:** the excess is positive at all 19 instances with `e = e_min`: n = 15, 17 at e = 3; n = 19-25 at e = 4; n = 27 at e = 5 (2 seeds).
- **No bug condition fired.**
- **Scope.** P is only testable at `e <= e_min`. Above `e_min` a positive excess is forced by `K_e <=(sub) K_{e+1}` (see the preregistration).

## Reading: what the extra kernel is `[pcc:unverified]`

Points 1-4 below are measured. Points 5-8 are post-hoc: two of them come with a short proof and one with an out-of-sample check. None of it is machine-checked.

1. **Degree profile.**
   - At `e = e_min`, every extra element has degree exactly `e_min`: `dim (K  cap  deg <= e_min - 1) = dim T` in every cell.
   - So the extra kernel is a set of new top-degree generators. Since `K_{e_min-1} = T_{e_min-1}`, none of them is a multiple of a lower-degree kernel element.
2. **Not V-independent and not field-structured.**
   - Take the subfamily with `lambda(C) perp Phi*_C(GF(2^n))` and `perp h_C` at every `C`. It equals `T_e` exactly in all 37 cells. The only V-independent syzygies are the trivial ones.
   - The subspace mapped back into the kernel by coefficientwise squaring (Frobenius) has dimension 0 in every cell, and so does the one for multiplication by `x`. This holds even for `T_e`, because `zeta` is fixed.
   - So the extra kernel is not a Frobenius orbit or a `GF(2^n)`-subspace.
3. **Support.**
   - The trivial family is exactly the set of kernel elements that vanish on `H_1` (in all cells).

- Every extra element is nonzero somewhere on H_1: rank_on_H1 = excess.

4. **Pointwise behaviour ("global generation"), measured in all 37 cells.**
   - At e = e_min, the evaluations of K_e at C span the whole pointwise annihilator A_C = W_C^perp, W_C = span(Phi*_C(V), h_C), at every one of the 2^k points.
   - At e = e_min - 1, they span zeta on H_0 and nothing on H_1.
   - So the extra syzygies appear at the same degree as global generation of the annihilator bundle.

5. **A recognisable sub-family on H_1: the mirror of the trivial family.**
   - A kernel element supported on H_1 is ell_0 * s, with s a section over H_1 of degree <= e - 1 in its k - 1 coordinates. So dim {lambda in K_e : lambda|H_0 = 0} equals the dimension of the H_1-sections at degree e - 1.
   - The data agree in every cell where both were computed:
     - n = 19, seeds 4 / 7 / 10: 105 / 108 / 105;
     - n = 27, seeds 3 / 6: 843 / 846;
     - 0 at n = 15, 17, 21, 23, 25.
   - So K_e = T_e (+) E_e (+) G_e:
     - T_e: supported on H_0, the F3 family;
     - E_e: supported on H_1, nonzero only at n = 19 and n = 27;
     - G_e: a complement of elements nonzero on both halves.
   - Sizes of G_e:
     - n = 19: 1164 - 130 - 105 = 929;
     - n = 27: 10400 - 1093 - 843 = 8464;
     - otherwise all of the excess.
   - The H_0-sections at degree e_min - 1 are exactly zeta * q. Their dimensions are M(k-1, e_min-1): 130 at n = 19, 232 at n = 23, 1093 at n = 27.

6. **Exact duality identity** (proved by hand below; checked numerically in 20 of 20 cells).

   nullity_e = n M(k,e) - (k+1) 2^k + Ann0 + dim S_{k-e-1}, where:

   - Ann0 = Sum_C (k+1 - rank W_C) is the dimension of the pointwise syzygies of the rows;
   - S_D = { G : V -> GF(2^n), deg G <= D, G(C) in W_C for all C }.

   Proof sketch:

   - The annihilator of the row space is {g : V -> F_2^{k+1} : deg Sum_i g_i w_i <= k-e-1}, by the duality between Reed-Muller codes of orders e and k-e-1.
   - The map g -> Sum g_i w_i has kernel the pointwise syzygies and image S_{k-e-1}.
   - With the obvious sections Sum g_i w_i, deg g <= D-1 (dimension (k+1)M(k,D-1) - M(k,D-2)), the identity becomes nullity = cols - E_eff + h + extra_S.
   - So the lead's E_eff is exactly this count with no extra sections. Its +1 is the point C = 0, where rank W_0 = k+1.

7. **The A/A variation is exactly h.**
   - Phi*_C(A) = A (A + C)(A^2 + CA + z), so rank W_C drops by one exactly when V contains a root of A^2 + CA + z. Then C = X1 + X2 with X1 X2 = z and X1, X2 in V. high_points.py checks that set equality on 19 instances, n = 15-27.
   - Hence Ann0 = 2^k - 1 + h, given sqrt(z) not in V, which holds in every instance.
   - The measured dim S_{k-e-1} at e_min does not depend on the instance: 800, 2486, 4070, 11936, 32034 at n = 15, 17, 19, 21, 23. So at e_min the kernel is a per-n base plus h.
   - At e_min - 1, dim S drops by exactly h: 1502 / 1501, 3836 / 3835, 6942 / 6941, 17712 / 17711. That cancellation keeps the nullity at M(k-1, e-1).
   - The out-of-sample prediction (n = 27, seed 6: 10401) held.
   - These h points are partial Semaev decompositions: pairs in V with the right product X1 X2 = z (not solutions of S = 0; the instances are unsatisfiable).

8. **Generic count against special sections.**
   - At n = 15 and 19, extra_S = 0: the kernel at e_min is exactly cols - E_eff + h (146 + h, 1163 + h). There the excess is a plain count of unknowns minus independent equations, not a special syzygy.

- At n = 17, 21, 23 there are sections of `W` beyond the obvious ones: 56, 210, 990 (measured). The identity gives 2640 at n = 25 and 3861 at n = 27, seed 3.
- 56 = C(8,3) and 210 = C(10,4), but 990 matches no such binomial. No closed form is claimed.

**Summary.** The extra syzygies at `e_min` consist of three parts: - an instance-dependent `+h`, explained by the curve-point structure (one per point `X1 + X2` with `X1 X2 = z`); - an `H_1`-supported mirror of the trivial family, when the `H_1`-sections at `e_min - 1` are nonzero; - a large global part that is no V-independent, Frobenius or `GF(2^n)`-linear family. At n = 15 and 19 that part is fully accounted for by the plain count, and elsewhere it is driven by extra low-degree sections of `W`.

## What is blocked or not done

- **`n = 25, 27` dual counts (`syz dual`) are not computed.** They need 177k and 349k columns, more than 2.2 GB, above the 3 GB / 20 min job budget. Their `extra_S` values come from the identity, not from a measurement.
- **`n = 29` and larger are not run** (`e_min` is 5 there): EKS/S3 are unavailable because the AWS identity is rejected. No resume checkpoint or bench record was written: no S3, and this is not a benchmark.
- **`research-kb` and `ecdb` were not queried** from this subagent. No novelty claim is made.
- **Not tested:** whether global generation (point 4) is equivalent to feasibility. It is the same degree in all 19 instances, but no argument links them. The planted and wrong-trace controls from Lead A1 were not run here either.

## Next step

1. Prove or refute: `e_min` = the least `e` at which `K_e` generates `A_C` at every point. Test it at the T-type instances (`V <=(sub) ker Tr`) and at `n = 29` on a larger box.
2. Identify the extra sections of `W` (56, 210, 990, ...), for example by checking whether they are `H_0`/`H_1`-supported sections of the dual bundle. A closed form for `dim S_{k-e-1}` would give `e_min` as a counting statement through the identity in point 6.
3. Lean obligations: the factorisation `Phi*_C(A) = A(A+C)(A^2+CA+z)`, and the duality identity (Reed-Muller duality plus rank-nullity).